

Order Orthoptera

Crickets and grasshoppers

With over 25,000 species in 40 families, crickets and grasshoppers are widespread in all but the coldest parts of the world. Most species have large wings—the front pair, or tegmina, being tough and leathery—but instead of flying away from danger, they often jump away on their powerful hindlegs. They have chewing mouthparts, but their diets vary: crickets, katydids, and their relatives (suborder Ensifera) are predators or omnivores, while grasshoppers and locusts (suborder Caelifera) are entirely vegetarian. All species undergo incomplete metamorphosis, and most adult males stridulate, or “sing,” to attract mates.

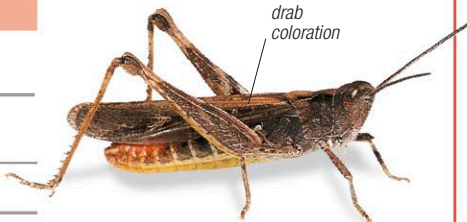
Family Acrididae

Grasshoppers

Occurrence 10,500 spp. worldwide; on ground, in vegetation



Grasshoppers are $\frac{3}{8}$ – $3\frac{1}{4}$ in (1–8 cm) long and have camouflage coloring and patterning. However, some have brightly colored bodies, or wing bands, to warn off predators. Some also produce noxious chemicals. The antennae are always short. Males sing during the day to attract mates. Females are nearly always larger than males and do not have a conspicuous ovipositor. After mating, they lay egg pods containing 15–100 eggs in the ground and protect them by secreting a foamy substance around them. Grasshoppers are agricultural pests; species that periodically form nomadic swarms—known as locusts—destroy crops in some parts of the world.

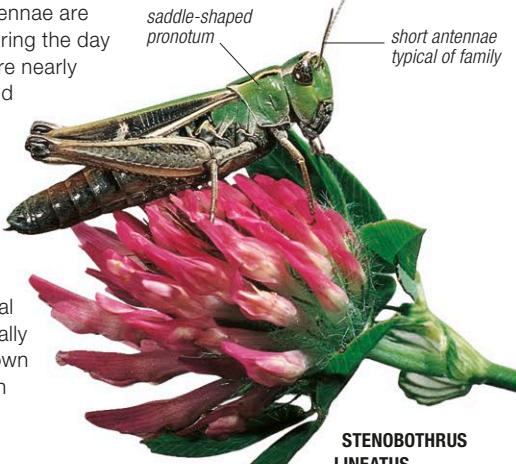


drab coloration

CHORTHIPPUS BRUNNEUS

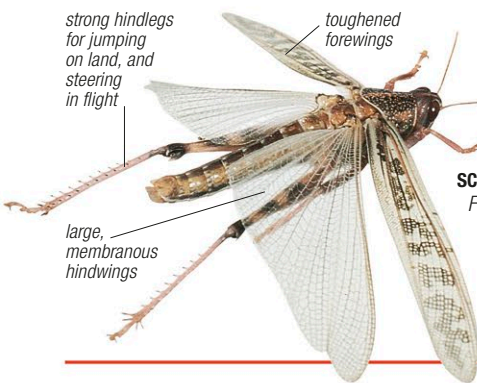
The common field grasshopper inhabits grassland from Northern Europe to Spain and Italy. It has a very distinctive song—a series of chirps repeated every 2 seconds or so.

saddle-shaped pronotum short antennae typical of family



STENOBOTHRUS LINEATUS

The stripe-winged grasshopper lives in dry, sunny grassland and pastures in Europe. It has a high-pitched, alternately loud and soft, song.



strong hindlegs for jumping on land, and steering in flight

toughened forewings

large, membranous hindwings

SCHISTOCERCA GREGARIA

Found from Africa to India, the desert locust is $2\frac{1}{4}$ in (6 cm) long. It has been a crop pest since agriculture first developed. Swarms of up to 50 billion individuals move with prevailing winds and can devour up to 110,231 tons (100,000 tonnes) of food in a day.

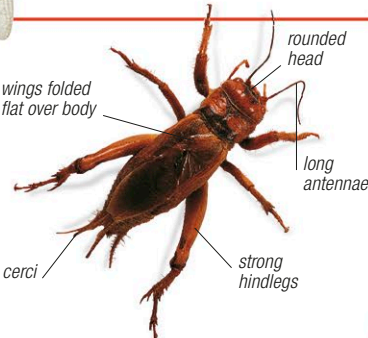
Family Gryllidae

True crickets

Occurrence 4,693 spp. worldwide; on ground, under stones, in leaf litter, on trees



True crickets are $\frac{3}{16}$ –2 in (0.5–5 cm) long, with slightly flattened bodies. Some species have loud, chirping songs, and are kept as pets. Most true crickets, except the carnivorous tree crickets, are either plant eaters or scavengers.



wings folded flat over body

rounded head

long antennae

strong hindlegs

cerci

BRACHYTRUPES SP.

Sometimes called tobacco crickets, these species are distributed across the African savanna. Relatives include pests that attack tea, tobacco, and cotton crops.

Family Gryllotalpidae

Mole crickets

Occurrence 105 spp. nearly worldwide; burrowing in damp sand or soil near streams, ponds, lakes



Like mammalian moles, these crickets are stout and covered with short, velvety hairs. Measuring $\frac{3}{4}$ – $1\frac{1}{4}$ in (2–4.5 cm) in length, they have short, broad, toothed front legs used for digging. Females dig chambers in the soil to lay their eggs; males make special burrows to amplify their songs, which can be heard up to 1 mile (1.5 km) away. Adults and nymphs eat parts of plants as well as small prey, feeding underground during the day and on the surface at night.



GRYLLOTALPA GRYLLOTALPA

The European mole cricket is attracted to lights after dark. It uses its front legs for digging, and hindlegs for pushing soil along burrows. It can be a crop pest.

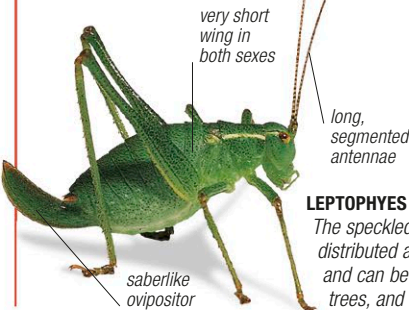
Family Tettigoniidae

Katydid

Occurrence 6,728 spp. worldwide, mainly in tropical regions; on vegetation



Also known as bush crickets, these insects get their name from a North American species whose song sounds like “kate-she did.” They measure $\frac{1}{2}$ –3 in (1.5–7.5 cm) in length and are brown or green, usually with big wings that slope over their sides. Most species mimic leaves or bark. They may also flash bright, hindwing colors to startle predators. Males sing using a file and scraper system at the base of the forewings. Females lay eggs in plants or soil. Most katydids feed and sing only at night.



very short wing in both sexes

long, segmented antennae

saberlike ovipositor

LEPTOPHYES PUNCTATISSIMA

The speckled bush cricket is distributed across Europe, and can be found in bushes, trees, and grassland.

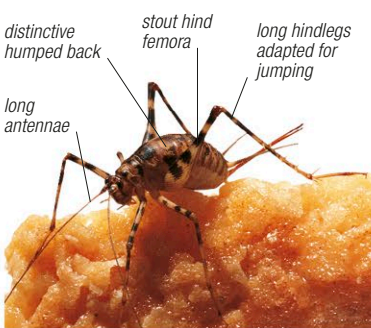
Family Rhaphidophoridae

Cave crickets

Occurrence 594 spp. widespread, especially in warmer regions; in caves, humid areas, under logs and stones

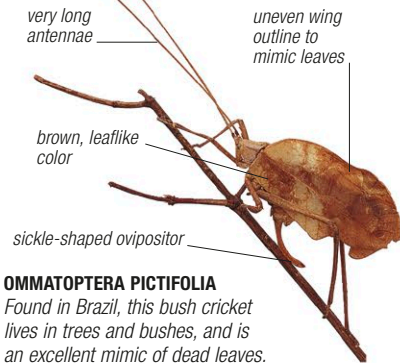


Squat, hump backed, and wingless, cave crickets are $\frac{1}{2}$ – $1\frac{1}{2}$ in (1.3–3.8 cm) long. They have long hindlegs and even longer antennae that help detect predators. These crickets are drab brown or gray in color. Some cave-adapted species have reduced eyes and soft bodies, and lay their eggs in cave-floor debris. Nymphs start searching for food as soon as they hatch; some scavenge, while others catch living prey.



PHAEOPHILACRIS GEERTSI

This African cave cricket is found only in the Democratic Republic of Congo.



very long antennae

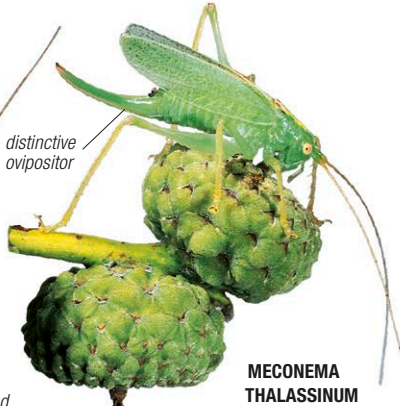
uneven wing outline to mimic leaves

brown, leaflike color

sickle-shaped ovipositor

OMMATOPTERA PICTIFOLIA

Found in Brazil, this bush cricket lives in trees and bushes, and is an excellent mimic of dead leaves.



distinctive ovipositor

MECONEMA THALASSINUM

The oak bush cricket can be found in Europe, mostly in oak woodland. It is mainly nocturnal and flies readily. Males signal to females by drumming their hindlegs against leaf surfaces.